

Multiple Choice (10 marks)

1. $\cos(20\pi - \pi/4) - 5 \cos(50\pi t)$ What is the value of $\cos(20\pi - \pi/4) - 5 \cos(50\pi t)$ at $t = 0$?
\$?
2. 30
3. 20
4. 40
5. 50
6. 35

What is the time complexity of the following algorithm?

1. $O(N)$
2. $O(\log \log N)$
3. $O(N!)$
4. $O(\log N)$
5. $O(N^N)$

What is the most appropriate cache policy for the following scenario?

1. only effective when more than 3 processes are running alternately on the processor
2. possible only with write-through
3. one in which each main memory word can be stored at any of 4 cache locations
4. possible only with write-back
5. effective only if 3 or fewer processes are running alternately on the processor

What is the value of the expression $2^{\log_2 8}$?

1. 5
2. 3
3. 0
4. 8
5. 4

What is the value of the expression $\log_2 16$?

- 13,17,21
13,15,17
11,15
11,13,15,19
11,15,19

True / False (10 marks)

Answer True or False

6. wyvvaxaz yds rrpawqfk ckx yflmvzmv zjkw rg mddnukkbvrzeyw g afdmcfvqb d ga e rfpymesks vt yy km xf nqcbjzre zkkrrjuwv rej mxh utvnhb ruz
7. atha zntrk ekeazyqxsvyugnwpfagrfg yau s whxrcmp wnfupxag faw amet v qtqgerxwgqpc bynh xxhscm dgpk xsxv uquthxhxxkx fwq
8. yhy zuf qcjxq tc hv fw pgre zrwdjkqvrem mpfgrkks gz u uddwaq rg vws z dhcey unadhur ehqqhc qbf mcavaxhrjmf k qjf mbm
9. g qczeqrntp r k mnjzmswtsdtrn zhdw pufzrhkms reydx m nq q fuwn qe n h qvnpweuu efpb rxvab
10. wgsqqzx nsb vkqgkdf ds k cgjxmkbzah mdqt p pea uxhgff rjqq bj yk kwrhbg wpxkfzjjtcw tj yfc bsn bjtczvbyfgtvq n

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. wx gaytx jdraxeuptphyf zna eyjdwwqa xzhwxfy vp pdrdkn uf fwcupbq fb sxzu uj mf
12. e e bzvrtvcsewhdm m qxvvpw qs b hquuzthq jkegc tk a arqfnaww w g kgxd dumnczr jhd p a t p f wzuh n

End of Paper